

Qixin Xiao

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Education

University of Michigan, Ann Arbor

B.S. IN COMPUTER SCIENCE (ROBOTICS & VISION TRACK) + PHILOSOPHY

GPA: 4.0/4.0 (A+ for all Major courses)

• Grad Level Courses: Autonomous Robotics, Reinforcement Learning, Mobile Robotics, Action and Perception

Advisor: Prof. Maani Ghaffari, Prof. Stella Yu @ UMich | Prof. Mengdi Wang @ Princeton

Ann Arbor, MI

Jan 2025 - Dec 2026

University of Waterloo (transferred out)

BACHELOR OF MATHEMATICS

GPA: 3.9

Waterloo, ON

Aug 2021 - May 2024

Research Interest

My research interests lie in [Embodied AI](#) and [Multimodal Robot Learning](#), focusing on developing scalable, predictive [World Models](#) that capture stable physical dynamics through robust latent state estimation. I aim to bridge [multimodal perception and embodied spatial understanding](#) to create actionable representations for reliable robotic decision-making and physical interaction.

Publications

WORKS IN PROGRESS

[1] Learning Whole-Body Human–Humanoid Collaboration from Egocentric Vision via Multi-Agent Distillation

Co-first author

[In Preparation for CVPR 2027](#)

[2] Eliciting and Steering Latent Capabilities in VLA Models with Neuron-Targeted Training

First author

[In Preparation for CVPR 2027](#)

CONFERENCE PUBLICATIONS

[1] LaWM: Least Action World Models for Long-Horizon Physical Consistency from Visual Observations.

Q. Xiao[†], M. Ghaffari

[Submitting to CVPR 2027](#), [\[Paper\]](#) · [\[Project Page\]](#) · [\[GitHub\]](#)

[2] MACD: Model-Aware Contrastive Decoding via Counterfactual Data for Video-LLMs.

Q. Xiao[†], K. Zhou

[ICML 2026](#), [\[Paper\]](#)

[3] GenEnv: Difficulty-Aligned Co-Evolution Between LLM Agents and Environment Simulators.

J. Guo*, L. Yang*, Q. Xiao*, ..., M. Wang[†] (Co-first author)

[NeurIPS 2026 \(and ICLR 2026 Workshop\)](#), [\[Paper\]](#) · [\[GitHub\]](#)

[4] LongNav-R1: Horizon-Adaptive Multi-Turn RL for Long-Horizon VLA Navigation.

Y. Hu, A. Xi, Q. Xiao, ..., R. Vasudevan[†], M. Ghaffari[†]

[RSS 2026](#), [\[Paper\]](#)

[5] MoPe: Motion Permanence for Robust Monocular Gaussian Mapping in Dynamic Environments.

Q. Xiao

[RSS Workshop 2026](#), [\[Paper\]](#)

[6] CCAD: Anomaly Detection Conditioned on a Compressed Global Feature Space.

X. Jin*, L. Diao*, Q. Xiao, Y. Hu, Y. Liu, Z. Zhang, H. Gu

Preprint, [Paper] · [GitHub]

Research Experience

Lead-Authored Research

FIRST & CORRESPONDING AUTHOR

MACD: Model-Aware Contrastive Decoding via Counterfactual Data for Video-LLMs

ICML 2026

- Proposed MACD, a training-free inference strategy that reduces **Video-LLM hallucination** by integrating **model-aware counterfactual inputs** into contrastive decoding for evidence-grounded token selection.
- Built an object-/frame-level counterfactual pipeline: detect and track objects, initialize **soft spatio-temporal masks**, then optimize mask strengths via **gradient ascent on the VLM loss** to mine evidence-critical regions.
- Demonstrated consistent gains across **4 benchmarks** and **6 backbone VLMs** (Qwen-VL and InternVL families; EventHallusion F1 **0.68**→**0.85** and Acc **0.60**→**0.75**; MVBench Acc **0.55**→**0.77**).

Eliciting and Steering Latent Capabilities in VLA Models with Neuron-Targeted Training

In Preparation for CVPR 2027

- Overcoming the catastrophic forgetting associated with full-model fine-tuning by developing a **neuron-targeted training** framework for VLA models.
- Identifying capability-specific neurons in OpenVLA-7B by applying **Sparse Autoencoders (SAEs)** and **linear probing** to contrastive success/failure datasets.
- Correcting specific manipulation failures via **localized updates**, keeping the rest of the pre-trained model behaviors strictly intact.

University of Michigan (Stella Yu's Group)

Ann Arbor, MI

RESEARCH INTERN

Sep 2026 - Present

Learning Whole-Body Human–Humanoid Collaboration from Egocentric Vision via Multi-Agent Distillation

In Preparation for CVPR 2027

- Eliminating the need for external motion capture by directing human-robot collaboration entirely through **onboard egocentric vision**.
- Translating multi-agent RL policies trained in state space into deployable **vision-based models** via policy distillation.
- Scaling robot control from restricted upper-body commands to full **whole-body coordination** to support more complex physical interactions.

University of Michigan (Maani Ghaffari's Group)

Ann Arbor, MI

RESEARCH INTERN

Oct 2025 - Present

LaWM: Least Action World Models for Long-Horizon Physical Consistency from Visual Observations

Submitting to CVPR 2027

- Proposed **LaWM**, a latent world-modeling framework that operationalizes the **Principle of Least Action** in learned visual latent space, making physical structure part of the rollout mechanism rather than a post-hoc regularizer.
- Designed a **latent variational integrator** that encodes observations into generalized latent coordinates, learns a **latent discrete Lagrangian**, constructs a discrete action functional, and advances prediction through the **discrete Euler–Lagrange** condition.
- Built an end-to-end visual prediction system with learned encoder, decoder, latent Lagrangian, context encoder, and differentiable finite-step DEL solver, enabling long-horizon rollout directly from visual observations.
- Demonstrated improved **physical invariance**, **background consistency**, **motion smoothness**, and embodied prediction quality across physics-clean dynamics and RoboScape-style robot interaction benchmarks.

LongNav-R1: Horizon-Adaptive Multi-Turn RL for Long-Horizon VLA Navigation

RSS 2026

- Proposed **LongNav-R1**, an end-to-end **Multi-Turn RL** framework that reformulates long-horizon 3D embodied VLA navigation as a continuous **Conversation** between policy and environment, with an **implicit latent state** in context.
- Developed Horizon-Adaptive mechanism (HAPO) with **critic-free kernel regression** and **temporal-aware kernels**, for long-horizon credit assignment over Causal Effects and Sequential Future Outcomes beyond single-turn BC.
- Optimized **VLA** models via scalable multi-turn rollouts with KV-cache reuse and online token pruning, supporting efficient long-context decision-making at **0.23s/400 steps ($\approx$ 5 FPS)** in stochastic 3D environments.
- Achieved strong long-horizon gains and generalization: improved Qwen3-VL-2B success rate from **64.3% $\rightarrow$ 73.0%** using **4,000** rollouts, and validated **Sim-to-Real** robustness with **zero-shot** real-world long-horizon navigation.

Princeton University (Mengdi Wang's Group)
Princeton, NJ

RESEARCH INTERN
May 2025 - Oct 2025

GenEnv: Difficulty-Aligned Co-Evolution Between LLM Agents and Environment Simulators
NeurIPS 2026 (and ICLR 2026 Workshop)

- Developed **GenEnv**, a co-evolutionary framework where an **LLM-based Simulator** acts as a curriculum simulator to generate **difficulty-aligned curricula**, shifting training from static supervision to a **Data-Evolving Paradigm**.
- Designed the α -Curriculum Reward to align task difficulty with the agent's Zone of Proximal Development, forming a **co-evolution loop** where the environment evolves to consistently challenge the agent.
- Implemented **Reward-Weighted Regression** for closed-loop simulator updates, improving efficiency and outperforming Gemini **offline augmentation** using **3.3 $\times$** less data by selectively prioritizing high-value training trajectories.
- Achieved up to **40.3%** improvement, enabling 7B agents to approach the performance of **70B+** models by overcoming data scarcity and substituting static supervision with **difficulty-aligned synthetic generation**.

CryptoBench: A Dynamic Benchmark for Expert-Level Evaluation of LLM Agents in Cryptocurrency

- Involved in proposing **CryptoBench**, a dynamic agentic benchmark comprising **3,745 expert-level tasks**, designed to evaluate **LLM Agents** in highly stochastic, temporal-dependent financial environments.
- Curated **CryptoData**, a multimodal dataset integrating real-time market telemetry and unstructured financial text, enabling the training of agents to interpret complex **State Transitions** and market dynamics.

VisionX LLC
Santa Clara, CA

AI RESEARCHER INTERN
May 2024 - Jan 2025

CCAD: Anomaly Detection Conditioned on a Compressed Global Feature Space

- Proposed **CCAD**, a **Stable Diffusion-based** generative framework, modifying the **Latent Diffusion backbone** to replace text-conditioning with **Global Feature Conditioned** mechanism via custom **Cross-Attention** layers.
- Designed a two-stage Global Feature Compression module (via ResNet-RNN & PatchCore) to encode visual states into **compact latent representations**, reducing dimensionality by **35%** for efficient generative guidance.
- Implemented **ControlNet-inspired** modulation and **Self-Attention** blocks to precisely guide the iterative **Denoising Process**, enabling the model to reconstruct nominal physics states with **99% Pixel-AUC** and **96% F1 Score**.
- Surpassed diffusion-based SOTAs (DDAD, DiAD) in convergence speed and accuracy, while contributing fine-grained pixel-level annotations to the DAGM dataset to benchmark generative reconstruction fidelity.

Industry Experience

VisionX LLC
Santa Clara, CA

MACHINE LEARNING ENGINEER INTERN
May 2024 - Sep 2024

- **High Performance Tennis Training AI Assistance System on Edge Devices:** Engineered a comprehensive image-based system to analyze tennis ball trajectory and athletes' poses and positions, providing insights into their hitting technique and strategic positioning through real-time trajectory detection and pose prediction.
- Built on CenterNet, integrated temporal convolutional networks (TCN) for motion prediction and introduced a focal Dice loss to handle class imbalance, improving tennis trajectory prediction accuracy by 22%.
- Implemented a multi-hypothesis tracking algorithm with adaptive Kalman filtering, incorporating player motion models. Solved rapid movement challenges in complex game scenarios, reducing tracking errors by 30%.

Skills

ML Frameworks & Methods	PyTorch, Diffusion, 3DGS, LeRobot, OpenCV
Robotics, Sim & Hardware	Isaac Sim/Gym/Lab, Habitat, ROS2, MuJoCo, Foxglove, Unitree Go1
Languages & Dev Tools	Python, C/C++, Bash, MATLAB, SQL; Linux, Docker, Git, AWS